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RESEARCH ARTICLE

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Intervention components, training dose, and adherence in exercise-based prevention of hamstring strain injury in football: a systematic review and meta-analysis

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ABSTRACT

Objective: To quantify associations between exercise-based prevention programmes and hamstring strain injury (HSI) risk in football participants, and whether training dose and adherence modify effects.

Methods: Six databases were searched to 1 October 2025. Randomised and cluster-randomised trials comparing HSI prevention programmes with usual practice or warm-up in football participants were included. Random-effects meta-analysis pooled risk ratios (RRs); subgroup analyses and meta-regression assessed effect modification.

Results: Fifteen trials (n = 7,465) were analysed. Programmes reduced HSI risk (RR = 0.51, 95% CI 0.36–0.71), with I^2=57% and a prediction interval crossing the null (0.18–1.40). Based on a control event rate of 7.8%, absolute risk reduction was 3.8% (38 fewer HSIs per 1000 participants; 95% CI 23–50 fewer). Effects were stronger for shorter interventions (1–6 months; RR = 0.43) than longer interventions (7–10 months; RR = 0.77; P for interaction=0.04), and for elite/semi-professional players (RR = 0.38) than amateur players (RR = 0.77; P for interaction = 0.02). Training frequency and weekly volume did not modify effects, whereas adherence did. High adherence (≥75%) was associated with lower HSI risk (RR = 0.36, 95% CI 0.28–0.48), whereas low adherence (<75%) showed no clear benefit (RR = 0.92, 95% CI 0.68–1.23; P for interaction <0.00001). Each 10% increase in adherence corresponded to an RR multiplier of 0.83 (approximately 17% lower RR). Certainty of evidence was low.

Conclusion: Exercise-based programmes reduce HSI risk in football when implementation supports sustained adherence. Effects may be stronger in shorter interventions and elite populations, but evidence remains insufficient to differentiate programme types or components.

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KEYWORDS
Hamstring injury; injury prevention; adherence; football; meta-analysis

1. Introduction

Hamstring strain injury (HSI) is among the most frequent non-contact muscle injuries in football (association football; soccer) and typically occurs during sprinting, rapid deceleration, and cutting actions [1,2]. HSIs represent a major burden in professional football, accounting for substantial absence days (approximately 25% of all injury-related absences) and requiring meaningful time loss for return to play (mean 9 days) [3]. Incidence estimates vary across cohorts and settings, with reports ranging from 1.06 to 5.87 injuries per 1000 h of exposure [4]. Surveillance in elite European teams indicates that a greater proportion of HSIs occur during match play than training, underscoring their close association with high-intensity game demands [5]. Beyond availability, HSIs impose substantial competitive and economic costs in elite football [3,6].

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Recurrence rates of 12–16% within the first year further compound the burden [7,8]. Collectively, these factors position HSI prevention as a practical priority in applied football settings [9–11].

Additionally, exercise-based prevention programmes represent the primary approach to HSI prevention. Preventive strategies evaluated in football include eccentric strengthening protocols such as the Nordic hamstring exercise (NHE), structured warm-up programmes such as FIFA 11/11+, and multicomponent programmes incorporating core stability and balance training [12,13]. Meta-analytic evidence generally supports risk reduction when the NHE is included within prevention programmes, with pooled estimates indicating that hamstring injury rates may be reduced by approximately half [14]. Evidence syntheses also suggest that eccentric resistance training and FIFA 11+ can decrease HSI occurrence in athletic populations, with some meta-analyses reporting reductions of up to 50% [15,16]. However, translation into football practice is complicated by variability in programme content (isolated eccentric training vs multicomponent programmes), prescription (frequency, volume, progression), delivery (integration into routine practice, supervision), and adherence [17–19].

Moreover, HSIs are commonly linked to repeated eccentric contractions and high muscular tension demands, supporting eccentric strengthening as a logical preventive target [19,20]. In parallel, modifiable factors such as strength and muscle architecture have been highlighted as key targets for prevention-oriented programming [21,22]. Multicomponent programmes may additionally influence HSI risk by improving movement control and load management during high-speed running and directional changes [23]. Therefore, the present study examines whether preventive effects differ by programme type, constituent elements, training dose, and adherence [9].

Despite extensive research, key implementation limitations remain. High-level teams may not adopt HSI prevention protocols consistently, and delayed-onset muscle soreness (DOMS) is commonly cited as a perceived barrier to the uptake of eccentric-focused exercises such as the NHE, although direct empirical evidence linking DOMS to reduced adherence in trials remains limited [17,24,25]. Furthermore, adherence appears to meaningfully influence effectiveness, yet has not been consistently quantified across trials and reviews [14,15,18]. For example, fewer than 15% of teams have been reported to complete the recommended FIFA 11+ volume, which may compromise real-world effectiveness [26]. Additionally, methodological critiques have questioned the certainty of some large effect estimates, suggesting that recommendations may be conditional on study design and analytic choices [27]. These issues collectively reinforce the need for synthesis approaches that explicitly account for programme modality and implementation factors, particularly adherence and dose.

Accordingly, the primary objective of this systematic review and meta-analysis was to quantify the association between exercise-based hamstring injury prevention programmes and hamstring strain injury (HSI) risk in football participants. Secondary objectives were to assess whether prescribed training dose, adherence, and playing level modify programme effects. Exploratory analyses compared intervention types, specific programme components, and programme complexity, and meta-regression quantified the adherence–effect association.

2. Methods

2.1. Methodology and literature search strategy

This meta-analysis was registered in PROSPERO (CRD420251246295) and reported in accordance with the PRISMA 2020 guidelines [28]. A completed PRISMA 2020 checklist is provided as Supplementary File 1. A comprehensive literature search was conducted in PubMed, Web of Science, Scopus, the Cochrane Library, EBSCOhost-SportDiscus, and EMBASE from database inception. The final search was completed on 1 October 2025. Searches were limited to full-text, peer-reviewed journal articles published in English. The search strategy was developed based on the PICOS (Population, Intervention, Comparison, Outcomes, Study design) framework [29], and the search strategies are provided in the (Supplementary Material Table S1), with any remaining database-specific strategies available on request. Additionally, reference lists of relevant articles were hand-searched to identify further eligible studies.

2.2. Inclusion and exclusion criteria

Randomised controlled or cluster-randomised trials in football participants and other football-related cohorts were eligible if they compared an exercise-based hamstring injury prevention programme with usual practice or a standard warm-up, and reported HSI events as event counts. The referee trial was included; its influence was assessed in sensitivity analyses. HSI was extracted as defined in each trial and analysed as total HSI events (including first-time and recurrent injuries when reported). Usual practice/standard warm-up referred to the team's routine training and warm-up activities without an additional structured HSI prevention programme. Studies enrolling mixed populations were eligible if football-specific HSI data could be extracted separately.

Furthermore, studies were excluded if they used non-randomised designs, evaluated rehabilitation or post-surgical interventions, did not report HSI-specific outcomes (or did not allow extraction of football-specific HSI data), involved non-football cohorts without separable data, or were not available as full-text articles. Reviews, case reports, and animal studies were also excluded.

2.3. Data extraction

Two reviewers independently extracted data using a prespecified form and entered them into a standardised spreadsheet. Data extraction covered study and participant characteristics (country, sex, age, and playing level) and intervention characteristics (type, duration, session frequency, session duration, and adherence). The population variable distinguished football participants from other football-related cohorts, such as referees. Intervention content was further coded according to predefined components (strength, balance, plyometrics, agility/speed, and flexibility) and by whether the programme incorporated eccentric hamstring training. Then, the programme complexity was classified as single- or dual-component versus multicomponent (≥3 components). Discrepancies were resolved by consensus and, when necessary, by adjudication from a third author. The complete extracted dataset is provided in the online supplementary material.

2.4. Quality assessment

Methodological quality was evaluated using the Cochrane Risk of Bias 2 (RoB 2) tool [30], covering five domains: the randomisation process, departures from intended interventions, missing outcome data, outcome measurement, and selection of the reported result. Assessments were completed independently by two reviewers. Discrepancies were resolved by consensus and, when necessary, by adjudication from a third author. RoB 2 was applied at the outcome level (HSI) for each trial. Overall risk of bias for each study was determined according to the RoB 2 algorithm (low risk, some concerns, or high risk).

2.5. Certainty of the evidence

The certainty of evidence was assessed using the GRADE (Grading of Recommendations, Assessment, Development and Evaluation) framework [31], considering risk of bias, inconsistency, indirectness, imprecision, and publication bias, with findings rated as high, moderate, low, or very low certainty.

2.6. Data analysis

Effect sizes were expressed as risk ratios (RRs) with 95% confidence intervals (CIs) and analysed on the log scale. For each study, log RRs and corresponding standard errors were obtained or derived from the reported data in accordance with the Cochrane Handbook for Systematic Reviews of Interventions [32]. For studies reporting injury incidence rates, event counts were extracted when available to enable consistent RR estimation. For cluster-randomised trials, we used effect estimates as reported in the original publications when available; otherwise, risk ratios were derived from the reported event counts at the study level. For studies with zero events in one arm, a continuity correction of 0.5 was applied to enable

estimation of risk ratios. Pooled estimates were obtained using inverse-variance weighting under a random-effects model [33]. Statistical heterogeneity was evaluated using Cochran's Q and quantified with the I^2 statistic. Subsequently, between-study variance (τ^2) was estimated, and prediction intervals for the pooled effect were calculated.

Prespecified subgroup analyses examined whether effects differed by training dose (intervention period, session frequency, session duration, and weekly training volume), adherence (low <75% vs high ≥75%), and playing level. High adherence was defined as completing ≥75% of prescribed training sessions. Adherence information was extracted from trial reports and standardised as the percentage of prescribed sessions completed (relative to the prescribed number) for analysis at the study level in our subgroup and meta-regression analyses. Exploratory subgroup analyses assessed intervention type, specific programme components (strength, balance, plyometrics, agility/speed, flexibility, and eccentric hamstring training), and programme complexity. Programme complexity was defined a priori as single- or dual-component versus multicomponent, with multicomponent programmes comprising three or more components. Studies were excluded from adherence-based analyses when adherence data were unavailable.

Random-effects meta-regression was conducted to examine the association between adherence (%) and intervention effects on the logRR scale. Sensitivity analyses assessed robustness using leave-one-out procedures and re-estimation under a fixed-effect model. In addition, the primary meta-analysis was repeated excluding studies judged to be at high risk of bias overall. Publication bias was assessed visually using funnel plots and statistically using Egger's regression and Begg's rank correlation tests when at least ten studies were available [34,35]. When asymmetry was suggested, the trim-and-fill method was applied.

All analyses were conducted in R (v4.4.3) in RStudio, primarily using the meta and metafor packages.

3. Results

3.1. Literature search results

A systematic search of six databases (PubMed, Web of Science, Scopus, EMBASE, Cochrane Library, and EBSCOhost-SportDiscus) identified 1,481 records. After removing 449 duplicates, 1,032 records were screened at the title and abstract level, of which 968 were excluded (968/1,032; 93.8%). Sixty-four full-text reports were sought; seven could not be obtained (7/64; 10.9%), leaving 57 for eligibility assessment. At full-text review, 46 reports were excluded (46/57; 80.7%) due to no injury outcomes reported (19/46; 41.3%), insufficient data for analysis (17/46; 37.0%), or ineligible study design (10/46; 21.7%). Citation searching identified four additional records, all of which met the eligibility criteria (4/4; 100%). In total, 15 studies were included in the systematic review and meta-analysis (Figure 1).

3.2. Characteristics of included studies

Fifteen trials were included (Table 1), enrolling football participants aged 13–45 years across ten countries [10,36–50]. Eleven studies included male-only cohorts, and four included female-only cohorts. Competitive level ranged from high school and amateur to semi-professional, professional, collegiate, and elite teams; one study enrolled amateur referees, whereas the remaining trials enrolled football players.

Furthermore, the interventions were exercise-based injury prevention protocols integrated into routine training. Programmes evaluated FIFA 11+ ($n = 4$), the NHE alone ($n = 4$), NHE with additional components ($n = 3$), and single trials of balance-board training, plyometric bounding, flywheel training, and a multimodal programme. Prescribed training dose ranged from 1.5 to 3.5 sessions/week (median 2) over 7–43 weeks (median 21). Session duration was reported in 11/15 trials (4–20 min), most commonly around 15 min. Adherence was reported in 14/15 trials, ranging from 21% to 100% (median 84%).

Across interventions (Table 2), strength elements were present in 14/15 trials and eccentric hamstring training in 12/15. The balance, plyometrics, and flexibility components each appeared in 6/15 trials, whereas the agility/speed components appeared in 5/15. By intervention type, FIFA 11+ accounted for 4/15 trials, NHE-focused protocols (including combinations) for 7/15, and other interventions for 4/15.

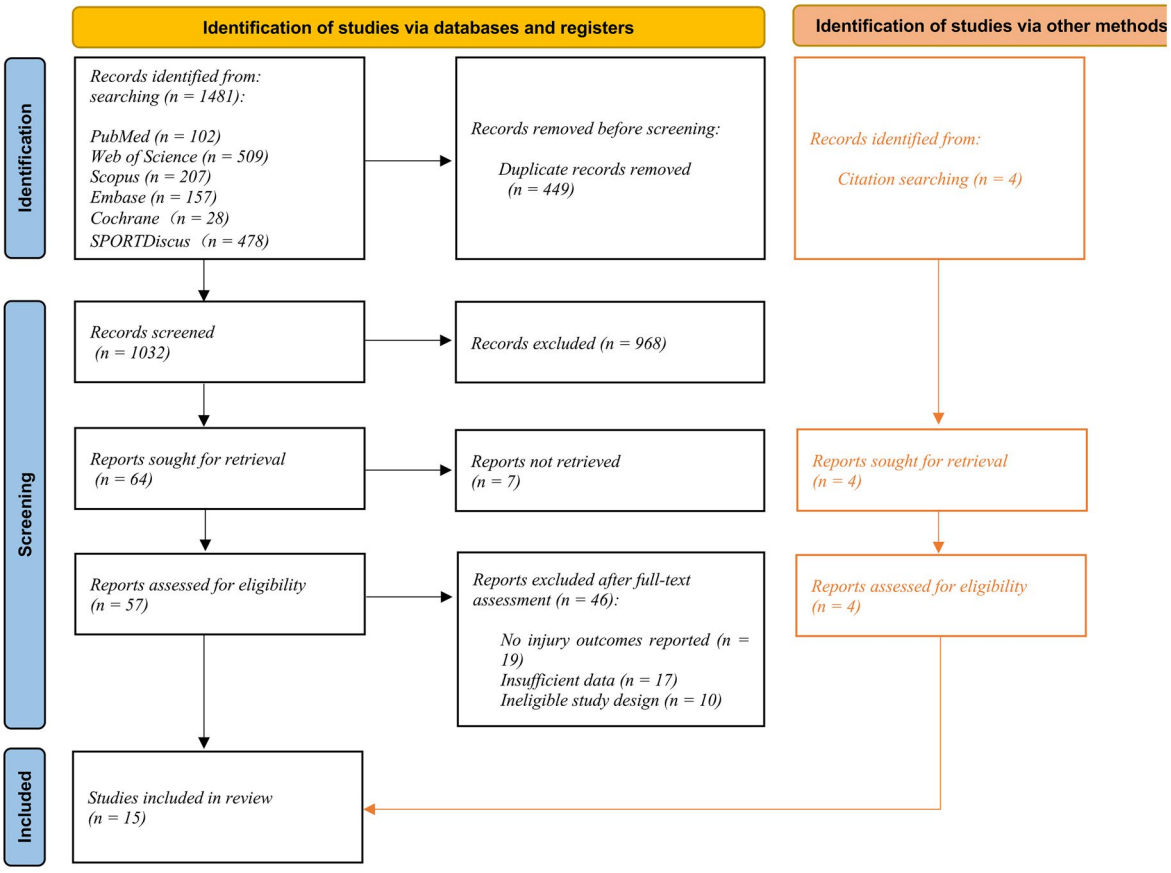


Figure 1. PRISMA flow diagram illustrating the study selection process for the systematic review and meta-analysis.

Most interventions were single- or dual-component (10/15), with multicomponent programmes (≥3 components) comprising 5/15.

3.3. Risk of bias assessment

Methodological quality for the 15 included studies was assessed using the Cochrane RoB 2 tool (Figure 2). Overall, one study (1/15) was judged to be at low risk of bias, twelve studies (12/15) raised some concerns, and two studies (2/15) were at high risk of bias. Domain-level assessment indicated that deviations from the intended interventions (D2) were the most common issue, raising some concerns in 14 out of 15 studies (14/15). Missing outcome data (D3) raised concerns in three studies (3/15), and selection of the reported result (D5) raised concerns in two studies (2/15). In contrast, the randomisation process (D1) demonstrated high methodological quality across most studies (14/15 at low risk), and measurement of the outcome (D4) was at low risk in most studies (13/15). These limitations, largely related to deviations from intended interventions, may have attenuated the estimated intervention effects.

3.4. Evaluation of meta-analysis results

3.4.1. Overall effect test

A meta-analysis of 15 randomised and cluster-randomised trials ($n = 7{,}465$) showed that exercise-based prevention programmes significantly reduced the risk of HSI in football participants (RR = 0.51, 95% CI 0.36–0.71; $p < 0.0001$), corresponding to an approximately 49% relative risk reduction (1 − RR). Between-study heterogeneity was moderate ($I^2 = 57\%$, $\tau^2 = 0.20$). The 95% prediction interval ranged from 0.18 to 1.40. It crossed the line of no effect (RR = 1), indicating that the magnitude of benefit might vary across implementation contexts. 14 of the 15 studies had point estimates favouring the intervention (RR < 1), supporting consistency in the direction of the overall effect (Figure 3).

Table 1. Data extraction from included articles.

Study	Country	Sex	Age (mean/range)	Athlete level	Intervention details
Al Attar et al. (2021)	Saudi Arabia	Male	31.6 ± 4.1	Male amateur soccer referees (lower leagues)	Adopted the FIFA 11+ program; intervention lasted 26 weeks, with ≥2 sessions/week and 20 min/session; overall adherence was 77.0%.
Askling et al. (2003)	Sweden	Male	24 ± 2.6	Elite male soccer players (Sweden's top premier-league teams)	Implemented the YoYo Flywheel program; 10-week intervention, 1–2 sessions/week, 15 min/session; adherence was 100.0%.
Elerian et al. (2019)	Egypt	Male	24.3 ± 3.8	Professional soccer players (sporting clubs in Alexandria, Egypt)	Intervention was the Nordic hamstring exercise (NHE); 12 weeks, 2 sessions/week, 20 min/session; adherence was 99.0%.
Engebretsen et al. (2008)	Norway	Male	17–35	Male soccer players (Norwegian 1st/2nd/top 3rd division teams)	Used NHE combined with balance training; 10 weeks, 1–3 sessions/week; session duration not reported; adherence was 21.0%.
Espinosa et al. (2015)	Spain	Female	21 (19–23)	Women's soccer players (Spain's 1st/2nd division teams)	NHE combined with EBE training; 21 weeks, 2 sessions/week, 15 min/session; adherence was 80.0%.
Hasebe et al. (2020)	Japan	Male	15–18	Male high school / soccer club players (top 16 schools in Saitama Prefecture league)	Implemented NHE; 27 weeks, 1–2 sessions/week; session duration not reported; adherence was 88.0%.
Heidt et al. (2000)	USA	Female	14–18	Female high school soccer players aged 14–18 years	Used the Frappier multimodal program; 7 weeks, 3 sessions/week; session duration not reported; adherence was 100.0%.
Petersen et al. (2011)	Denmark	Male	23.5	Danish male soccer players (professional/semiprofessional/amateur; divisions 1–5)	Intervention was NHE; 10 weeks, 1–3 sessions/week, 5–10 min/session; adherence was 91.0%.
Sebelien et al. (2014)	Norway	Male	18–39	Male semi-professional soccer players aged 18–39 years (Norway level 3/4)	Used NHE combined with stretching; 43 weeks, 3 sessions/week; session duration not reported; adherence not reported.
Silvers-Granelli et al. (2015)	USA	Male	18–25	Male collegiate soccer players (US NCAA Division I/II)	Implemented FIFA 11+; 18 weeks, 2–3 sessions/week, 20 min/session; adherence was 91.0%.
Soligard et al. (2008)	Norway	Female	13–17	Female youth soccer players aged 13–17 years (125 Norwegian clubs)	Used FIFA 11+ ('The 11+'); 34 weeks, 2–5 sessions/week, 20 min/session; adherence was 77.0%.
Söderman et al. (2000)	Sweden	Female	20.4	Female soccer players (Sweden's 2nd/3rd division teams)	Intervention was balance-board training; 30 weeks, 3 sessions/week, 10–15 min/session; adherence was 65.0%.
van de Hoef et al. (2019)	Netherlands	Male	18–45	Adult male amateur soccer players (32 first-class amateur league teams)	Implemented a bounding exercise program (BEP; plyometrics); 39 weeks, 2 sessions/week, 3–5 min/session; adherence was 71.0%.
Van der Horst et al. (2015)	Netherlands	Male	24.5	Male amateur soccer players (KNVB Eerste Klasse; mean age 24.5 ± 3.8 years)	Intervention was NHE; 13 weeks, 2 sessions/week, 5–10 min/session; adherence was 91.0%.
Whalan et al. (2019)	Australia	Male	23.8	Male semi-professional soccer players (25 clubs; Tier 2/3 leagues)	Used FIFA 11+; 28–34 weeks, ≥2 sessions/week, 20 min/session; adherence was 58.0%.

Data are presented as mean ± standard deviation or range where available. Adherence is expressed as a percentage of sessions completed relative to the prescribed number, unless otherwise noted.

Abbreviations: NHE, Nordic Hamstring Exercise; EBE, Eccentric Band Exercise; BEP, Bounding Exercise Program; FIFA 11+, Fédération Internationale de Football Association 11+ injury prevention program.

Table 2. Overview of intervention components and complexity in exercise-based prevention programs.

Study	Strength	Balance	Plyometrics	Agility/ Speed	Flexibility	Eccentric Hamstring Training	Intervention Program	Complexity
Al Attar et al. (2021)	Yes	Yes	Yes	Yes	Yes	Yes	FIFA 11+ Referees	Multi-component (≥3)
Askling et al. (2003)	Yes	No	No	No	No	Yes	YoYo Flywheel	Single/Dual-component
Elerian et al. (2019)	Yes	No	No	No	No	Yes	NHE	Single/Dual-component
Engebretsen et al. (2008)	Yes	Yes	No	No	No	Yes	NHE + Balance	Single/Dual-component
Espinosa et al. (2015)	Yes	No	No	No	No	Yes	NHE + EBE	Single/Dual-component
Hasebe et al. (2020)	Yes	No	No	No	No	Yes	NHE	Single/Dual-component
Heidt et al. (2000)	Yes	No	Yes	Yes	Yes	No	Frappier (Multi-modal)	Multi-component (≥3)
Petersen et al. (2011)	Yes	No	No	No	No	Yes	NHE	Single/Dual-component
Sebelien et al. (2014)	Yes	No	No	No	Yes	Yes	NHE + Stretching	Single/Dual-component
Silvers-Granelli et al. (2015)	Yes	Yes	Yes	Yes	Yes	Yes	FIFA 11+	Multi-component (≥3)
Soligard et al. (2008)	Yes	Yes	Yes	Yes	Yes	Yes	FIFA 11+ ('The 11+')	Multi-component (≥3)
Söderman et al. (2000)	No	Yes	No	No	No	No	Balance Board	Single/Dual-component
van de Hoef et al. (2019)	Yes	No	Yes	No	No	No	BEP (Bounding)	Single/Dual-component
Van der Horst et al. (2015)	Yes	No	No	No	No	Yes	NHE	Single/Dual-component
Whalan et al. (2019)	Yes	Yes	Yes	Yes	Yes	Yes	FIFA 11+	Multi-component (≥3)
Proportion (Count/ Total)	Yes: 14 No: 1	Yes: 6 No: 9	Yes: 6 No: 9	Yes: 5 No: 10	Yes: 6 No: 9	Yes: 12 No: 3	FIFA 11+: 4 NHE Group (incl. combinations): 7 Other Interventions: 4	Single/Dual-component: 10 Multi-component: 5

Note. 'Yes' indicates that the component was included in the intervention; 'No' indicates that it was not. Interventions were classified based on their complexity: Multi-component, comprising three or more training components (strength, balance, plyometrics, agility/speed, or flexibility); and Single/Dual-component, comprising one or two training components. NHE, Nordic Hamstring Exercise; BEP, Bounding Exercise Program; EBE, Eccentric Band Exercise; FIFA 11+, a comprehensive warm-up program designed by FIFA. 'Proportion' summarizes the frequency of each component or category across the included studies.

3.4.2. Subgroup analyses stratified by training dose, adherence, and playing level

There was no evidence of interaction by dose-related parameters, including training frequency, session duration, and weekly training volume (P for interaction = 0.54, 0.72, and 0.62, respectively; Supplementary Figure S1). Effect estimates consistently favoured the intervention across dose strata. Effects were broadly comparable for 1–2 sessions/week ($k = 8$; RR = 0.56, 95% CI 0.38–0.83) and 3–4 sessions/week ($k = 7$; RR = 0.45, 95% CI 0.24–0.84). Similarly, both 1–30 min/week ($k = 6$; RR = 0.42, 95% CI 0.26–0.68) and 30–60 min/week ($k = 4$; RR = 0.51, 95% CI 0.27–0.96) were associated with a lower HSI risk.

In contrast, there was evidence of subgroup differences by intervention duration, adherence, and playing level (P for interaction = 0.04, <0.00001, and 0.02, respectively). Interventions lasting 1–6 months ($k = 9$) were associated with a reduced HSI risk (RR = 0.43, 95% CI 0.28–0.66; $p = 0.0001$), whereas the 7- to 10-month subgroup ($k = 6$) did not reach statistical significance (RR = 0.77, 95% CI 0.54–1.09; $p = 0.14$).

Adherence showed the strongest evidence of effect modification. High adherence (≥75%; $k = 10$) was associated with a substantial reduction in HSI risk (RR = 0.36, 95% CI 0.28–0.48; $p < 0.00001$), whereas low adherence (<75%; $k = 4$) showed no clear benefit (RR = 0.92, 95% CI 0.68–1.23; $p = 0.56$). According to playing level, the association appeared stronger in elite/semi-professional players ($k = 8$; RR = 0.38, 95% CI 0.23–0.61; $p < 0.0001$), whereas the amateur subgroup ($k = 7$) was not statistically significant (RR = 0.77, 95% CI 0.53–1.10; $p = 0.15$).

Given the pronounced effect modification by adherence, additional adherence-stratified analyses were conducted across populations and implementation contexts to assess consistency (Section 3.4.3). Other

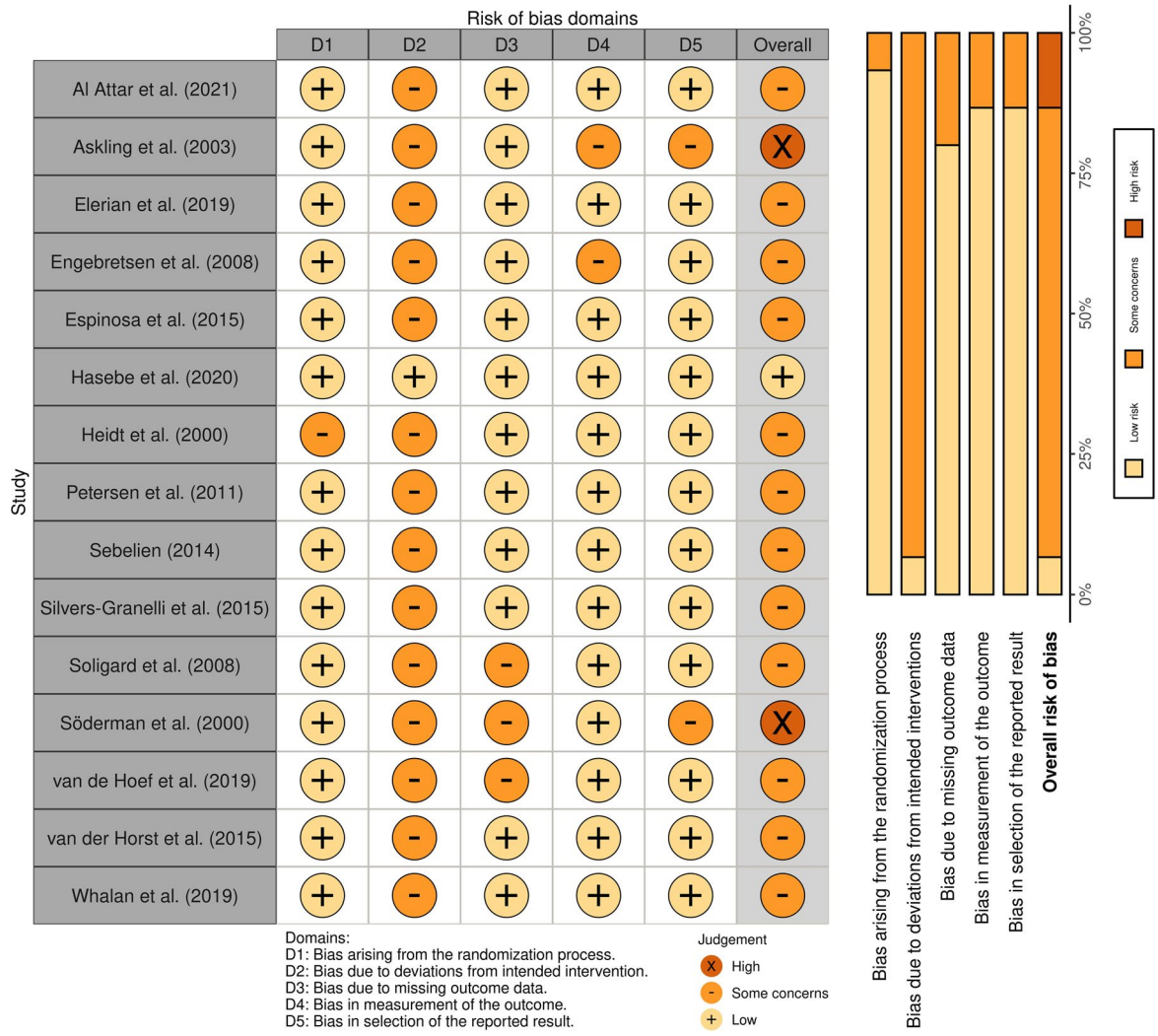


Figure 2. Risk of bias evaluation conducted in accordance with the Cochrane Collaboration guidelines.

subgroup findings, such as those related to intervention duration and playing level, were limited by multiple comparisons and small study numbers in some strata.

3.4.3. Adherence-stratified analyses

Adherence consistently modified intervention effects across populations and contexts (Supplementary Figure S2). Across strata, the pooled RR under low adherence (<75%) was close to 1 and non-significant: adults (RR = 0.92, 95% CI 0.68–1.23), amateur athletes (RR = 0.92, 95% CI 0.68–1.23), male athletes (RR = 0.94, 95% CI 0.72–1.22), and strength-based interventions (RR = 0.94, 95% CI 0.72–1.22). A similar pattern was observed for eccentric-based interventions: under low adherence (<75%), the pooled RR was 0.96 (95% CI 0.70–1.32), whereas under high adherence (≥75%), the pooled RR was 0.36, 95% CI 0.27–0.48. By contrast, high adherence (≥75%) was consistently associated with a substantial and statistically significant reduction in HSI risk across these strata (high-adherence pooled RRs clustered around 0.34–0.38: adults RR = 0.34, 0.25–0.46; amateurs RR = 0.38, 0.28–0.52; males RR = 0.35, 0.26–0.47; strength-based interventions RR = 0.36, 0.28–0.48; eccentric-based interventions RR = 0.36, 95% CI 0.27–0.48). The between-adherence subgroup differences were statistically significant within each stratum (P for interaction < 0.00001 throughout). Some strata might draw on overlapping sets of trials, which could yield identical or near-identical pooled estimates across related stratifications.

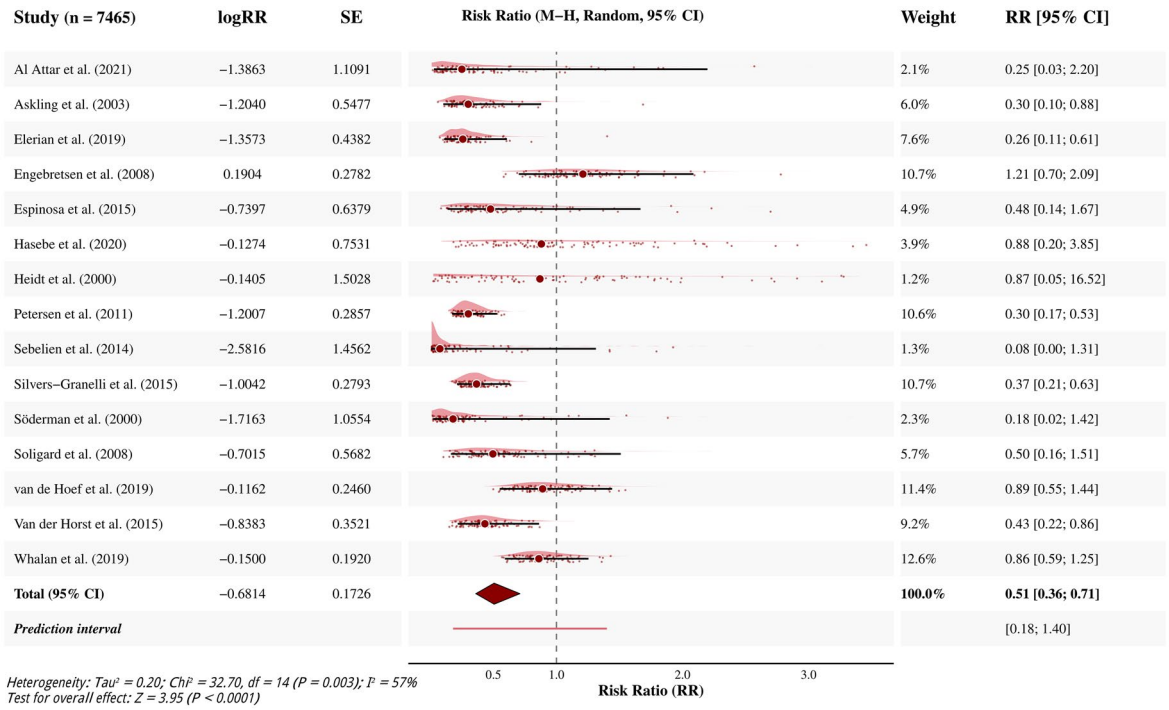


Figure 3. Pooled effect of exercise-based prevention programmes on football participants.

3.4.4. Exploratory subgroup analyses: intervention type, programme components, and complexity

Exploratory subgroup analyses examined whether intervention type and selected programme components (or programme complexity) were associated with differential effects on HSI risk. Point estimates generally favoured the intervention across strata. However, there was no evidence of subgroup differences for any comparison (all P for interaction > 0.05; Supplementary Figure S3). Given the exploratory nature of these analyses and the limited number of trials within several strata (as few as $k = 3$ in some comparisons, although the eccentric training stratum included $k = 12$), these findings should not be interpreted as evidence of equivalence between programmes.

By intervention type, FIFA 11+ (RR = 0.53, 95% CI 0.29–0.97; $k = 4$) and NHE-based programmes (RR = 0.46, 0.26–0.84; $k = 7$) were each associated with significant risk reductions, whereas other interventions were not (RR = 0.53, 0.24–1.18; $k = 4$); effects did not differ between intervention types (P for interaction = 0.94).

In addition, component-based analyses showed no evidence of subgroup differences by inclusion of balance training ($p = 0.37$), plyometrics ($p = 0.22$), agility/speed training ($p = 0.69$), flexibility training ($p = 0.97$), or programme complexity (single/dual vs multicomponent; $p = 0.69$). For eccentric training, the stratum including eccentric exercise showed a significant protective association (RR = 0.48, 0.33–0.70; $k = 12$), whereas the stratum without eccentric training did not (RR = 0.74, 0.36–1.53; $k = 3$). Nevertheless, the between-stratum difference was not statistically significant ($p = 0.28$).

3.4.5. Meta-regression: adherence as a predictor of intervention effects

One study did not report adherence; therefore, the meta-regression included 14 trials ($k = 14$). Random-effects meta-regression showed a significant inverse linear association between adherence (%) and intervention effects on the logRR scale (β = −0.0190, 95% CI −0.0272 to −0.0109; $p < 0.001$). This corresponded to an RR multiplier = 0.83 (95% CI 0.76–0.90) for each 10-percentage-point increase in adherence, equivalent to an approximate 17% reduction in RR (Supplementary Figure S4a). As this was an aggregate-level analysis, causal inference requires caution.

This dose–response association was consistent across prespecified subsets. Regression slopes were similar (β ranging from −0.0178 to −0.0194; all $p < 0.001$), corresponding to RR multipliers of 0.82–0.84

per 10-percentage-point increase in adherence: amateur athletes ($k = 11$; RR multiplier = 0.84, 95% CI 0.77–0.91), male athletes ($k = 10$; RR multiplier = 0.82, 95% CI 0.76–0.90), programmes including a strength component ($k = 13$; RR multiplier = 0.83, 95% CI 0.76–0.90), and programmes including eccentric training ($k = 11$; RR multiplier = 0.82, 95% CI 0.76–0.89; Supplementary Figure S4b–e). The close agreement across subsets was consistent with a stable adherence–effect dose–response relationship.

3.5. Sensitivity analyses of the pooled effect

Leave-one-out analyses showed consistent protective effects after omitting any single study (pooled RR range 0.33–0.64), with no single trial exerting undue influence on the overall estimate. A fixed-effect model yielded similar results (RR = 0.55, 95% CI 0.46–0.66). Results were robust to the exclusion of studies judged to be at high risk of bias (RR = 0.54, 95% CI 0.38–0.76). Excluding the study that enrolled referees did not materially change the pooled effect or heterogeneity (primary: RR = 0.51, 95% CI 0.36–0.71; I^2 = 57%; after exclusion: RR = 0.51, 95% CI 0.36–0.72; I^2 = 59%) [36]. Omitting Engebretsen et al. (2008) produced the largest reduction in heterogeneity (I^2 = 48%) while maintaining a significant protective effect (RR = 0.46, 95% CI 0.33–0.64).

3.6. Assessment of publication bias

Formal tests provided no evidence of funnel plot asymmetry. Neither Egger's regression test ($p = 0.09$) nor Begg's rank correlation test ($p = 0.66$) was statistically significant. Trim-and-fill analysis identified and imputed three potentially missing studies; the adjusted pooled estimate was slightly attenuated but remained in the protective direction (unadjusted: RR = 0.51, 95% CI 0.36–0.71; adjusted: RR = 0.54, 95% CI 0.39–0.75), suggesting that any small study effect, if present, would have limited impact on the overall conclusion (Figure 4). Overall, the available evidence does not suggest major publication bias, although these tests might be underpowered with a modest number of trials, and minor asymmetry could not be fully excluded.

3.7. GRADE assessment of the certainty of evidence

Certainty of the evidence was assessed using the GRADE framework. As all included studies were randomised trials, initial certainty was high but was downgraded by two levels to low. One level was downgraded for inconsistency due to moderate heterogeneity (I^2 = 57%) and a prediction interval that crossed the null, suggesting context-dependent effects. Another level was downgraded for risk of bias because most trials raised some concerns or were at high risk of bias overall, predominantly driven by deviations from intended interventions. No downgrading was applied for indirectness or imprecision because the

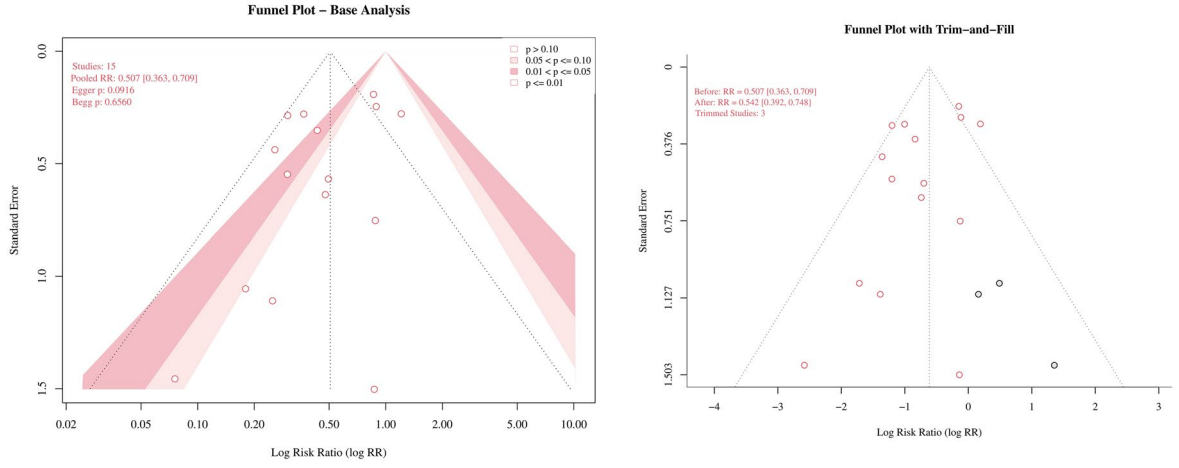


Figure 4. Funnel plot and trim-and-fill analysis assessing small study effects/publication bias for HSI.

evidence directly addressed the review question, and the pooled confidence interval excluded the null. Publication bias was not downgraded due to no clear evidence of asymmetry, acknowledging the limited power of these assessments.

4. Discussion

4.1. Main findings

This systematic review and meta-analysis of 15 randomised controlled trials found that exercise-based prevention programmes reduce the risk of HSI in football participants (RR = 0.51, 95% CI 0.36–0.71, $p < 0.0001$). By restricting inclusion to football-specific RCTs with HSI as the primary outcome, the present synthesis provides estimates with enhanced specificity for applied football settings.

The pooled effect exhibits moderate heterogeneity (I^2 = 57%), and the 95% prediction interval crosses the null (0.18–1.40), indicating that intervention effects may vary substantially across contexts. This variability suggests that real-world effectiveness is likely to differ depending on implementation conditions. Implementation conditions may include how programmes are integrated into routine training, the level of supervision and delivery fidelity, and the extent of sustained adherence and control-group contamination. When adherence is included as a covariate in meta-regression, residual heterogeneity is markedly attenuated (approaching zero), suggesting that variation in adherence may explain a substantial proportion of between-study differences.

The present study extends prior evidence by examining sources of heterogeneity rather than simply estimating average efficacy. Specifically, we quantified the adherence–effect gradient using meta-regression and examined prespecified effect modifiers (training dose, intervention duration, adherence, and playing level), alongside exploratory comparisons of programme type, components, and complexity. Both prespecified subgroup analyses and meta-regression indicate that study-level adherence was consistently associated with effect magnitude and demonstrate a dose–response pattern ($\beta = -0.0190$, $p < 0.001$). In contrast, training dose and programme type show weaker or inconsistent associations with outcomes. Accordingly, while exercise-based prevention programmes can reduce HSI risk in football, achieving meaningful benefits in practice may be contingent on sustained adherence and favourable implementation conditions. Meanwhile, evidence is insufficient to reliably differentiate programme types or prescribed training volumes within the ranges examined.

4.2. Training dose, adherence, and effect modifiers

The risk of bias assessment identifies deviations from intended interventions as the most common concern, suggesting that implementation factors such as adherence, contamination, or dose delivery may have attenuated between-group contrasts and contributed to heterogeneity. Similar implementation-related limitations have been noted in prior hamstring injury prevention reviews [14].

4.2.1. Training dose

Subgroup analyses by training frequency, session duration, and weekly volume show no statistically significant differences between dose strata, although point estimates generally favour protection across all dose levels. Specifically, interventions involving 1–2 sessions per week and 3–4 sessions per week are both associated with reduced HSI risk, with no clear advantage for higher-frequency (RR = 0.56 vs 0.45; P for interaction = 0.54). Similarly, weekly volumes of 1–30 and 30–60 min show comparable protective effects (RR = 0.42 vs 0.51; P for interaction = 0.62). A meta-analysis of lower limb injuries reported no effect for once-weekly hamstring eccentric training but a significant reduction for twice-weekly hamstring eccentric training (RR = 1.00 vs 0.60) [51], which may reflect differences in injury outcomes, participant populations, and intervention implementation across reviews. In contrast, in this study's football-specific HSI synthesis, both 1–2 and 3–4 sessions/week are similarly protective with no evidence of interaction, supporting the feasibility of lower-frequency prescriptions when adherence is maintained [52]. These findings suggest that, within the dose ranges examined, both lower and higher prescribed

doses may be effective provided they are implemented consistently. However, interpretation is limited by inconsistent definitions of dose metrics across trials and by potential discrepancies between the prescribed and completed dose, particularly in the context of low adherence. Consequently, given the limited statistical power due to small study numbers, the present data do not support strong inferences regarding optimal dose thresholds.

4.2.2. Intervention duration and playing level

Shorter interventions (1–6 months) demonstrate statistically significant effects (RR = 0.43, 95% CI 0.28–0.66), whereas longer interventions (7–10 months) do not reach statistical significance (RR = 0.77, 95% CI 0.54–1.09), with evidence of subgroup differences ($p = 0.04$). Similarly, elite and semi-professional populations show stronger effects (RR = 0.38, 95% CI 0.23–0.61) than amateur populations (RR = 0.77, 95% CI 0.53–1.10; P for interaction = 0.02). Because adherence emerged as a consistent effect modifier across multiple analyses, these duration and level differences may partly reflect systematic variation in adherence over time or across competitive contexts [17]. However, the specific mechanisms underlying these patterns are not directly examined in the included trials, and small study numbers within strata limit definitive interpretation.

4.2.3. Adherence

Adherence emerges as the most consistent effect modifier. High-adherence trials (≥75%) show substantial risk reduction (RR = 0.36, 95% CI 0.28–0.48), whereas low-adherence trials (<75%) cross the line of no effect (RR = 0.92, 95% CI 0.68–1.23; P for interaction < 0.00001). Meta-regression indicates a dose–response relationship between adherence and effect magnitude (β = −0.0190, $p < 0.001$), corresponding to an RR multiplier of 0.83 per 10-percentage-point increase in adherence. Practically, these findings suggest that meaningful HSI risk reduction is most likely when teams achieve and sustain high adherence (≥75% of prescribed sessions), with progressively smaller risk ratios as adherence increases. This pattern is consistent across prespecified subgroup analyses and across extended adherence-stratified analyses in different populations and implementation contexts.

Additionally, the findings align with broader evidence suggesting that injury prevention effectiveness increases with adherence [18,26,53]. Alternative explanations cannot be excluded. For example, trials achieving high adherence may also have differed in implementation fidelity, participant motivation, or control-group contamination. These findings should be interpreted as signals of effect modification and sources of heterogeneity rather than definitive causal evidence. Individual-level adherence-outcome data will be required to establish causality.

4.2.4. Intervention type and components

Exploratory comparisons of intervention types and programme components show no consistent differences. Given limited statistical power, small study numbers, and variability in intervention reporting and delivery, these findings should be interpreted as insufficient evidence to differentiate programme types rather than evidence of equivalence. Future trials should improve specification and reporting of both prescribed content and implementation processes, including adherence, fidelity, and progression, to enable more informative comparisons [27].

4.3. Implications for practice

The present findings suggest three considerations for implementing exercise-based HSI prevention programmes in football.

First, adherence appears to be an important determinant of intervention effectiveness. High-adherence trials demonstrate clear risk reduction, whereas low-adherence trials show attenuated effects [45]. Accordingly, routinely monitoring adherence and providing regular feedback within teams is likely a key implementation strategy to support sustained adherence and high-fidelity delivery. Programme selection

and delivery should therefore account for feasibility and sustainability within the team environment [18,53].

Second, training frequency and weekly volume in the ranges examined (1–4 sessions per week; 1–60 min per week) are both associated with protective effects, with no clear advantage for higher prescribed doses. This does not preclude differences by completed dose, which may deviate from the prescribed dose under variable adherence. These findings suggest that programmes can be effective across a range of dose levels, provided they are implemented consistently.

Third, shorter intervention durations and elite/semi-professional populations show stronger effects than longer interventions or amateur populations. Given that adherence is the most consistent effect modifier across analyses, these differences may partly reflect variation in adherence over time or across competitive levels [47].

4.4. Limitations

Several limitations should be considered when interpreting these findings.

First, moderate heterogeneity (I^2 = 57%) and a prediction interval crossing the null (0.18–1.40) suggest that effects may vary across contexts, and the pooled estimate reflects an average effect.

Second, dose and adherence are inconsistently defined and reported across trials, limiting comparability. Several included trials used cluster randomisation, and it was not always clear whether clustering was consistently accounted for in the original trial analyses; if clustering is not accounted for, precision may be overstated. In addition, adherence was extracted from trial reports and may vary in definition and measurement accuracy across studies, which could bias the estimated adherence–effect relationship. Subgroup and meta-regression analyses are conducted at the study level and are susceptible to ecological bias and confounding; adherence-outcome associations do not establish individual-level causality, and alternative explanations (such as differences in implementation fidelity or participant motivation) cannot be excluded. These analyses are therefore hypothesis-generating rather than causal. Third, subgroup analyses are based on small numbers of studies, and multiple comparisons increase the risk of chance findings; these exploratory results require confirmation.

Finally, generalizability is limited because most trials enrolled male players (11/15 male-only), and the small number of female-only trials (4/15) precluded sex-stratified analyses; extrapolation to female football, youth populations, and other team sports should, therefore, be cautious.

4.5. Future research

Several priorities for future research emerge from these findings. First, future trials should standardise reporting of training exposure and adherence using consistent definitions and core metrics, including session frequency, duration, progression, and attendance; basic process measures such as execution quality should also be documented to improve interpretability and enable cross-study comparisons. Second, comparative evaluations using head-to-head or factorial designs, with cluster randomisation where appropriate, will help clarify which programme components or combinations are associated with benefit, alongside monitoring of control-group activities to mitigate contamination. Third, adequately powered studies in female football and youth populations, together with pragmatic club-based implementation research, are needed to strengthen external validity and inform scalable delivery strategies.

5. Conclusions

Exercise-based hamstring injury prevention programmes reduce the risk of HSI in football participants. Adherence is consistently associated with effect magnitude across multiple stratifications, with preventive benefit concentrated in high-adherence implementations. Subgroup patterns also suggest differences by intervention duration and playing level. Within the examined prescribed-dose ranges, there was no clear additional advantage of higher-frequency versus lower-frequency prescriptions, nor of higher weekly training time versus lower weekly training time. Evidence remains insufficient to reliably rank programme

types or specific component configurations. Accordingly, programme selection should prioritise feasibility, and delivery should prioritise maintaining high adherence and implementation fidelity to maximise reproducible preventive gains.

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Not applicable. This study synthesised data from previously published research and did not involve new data collection from human participants.

Authors contributions

CRediT: **Jinfa Gu**: Conceptualization, Data curation, Formal analysis, Project administration, Visualization, Writing – original draft, Writing – review & editing; **Yan Li**: Data curation, Formal analysis, Software, Supervision; **Yinhao Wang**: Conceptualization, Investigation, Software; **Yu Zhang**: Formal analysis, Methodology, Validation.

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Data availability statement

The datasets analysed during the current study are available from the corresponding author on reasonable request. As this study is a systematic review and meta-analysis, all data were derived from published studies.

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